# SteppeDNA

Multi-Gene Pathogenicity Classifier for Homologous Recombination DNA Repair Genes

**Elnar Abzhakhanov**
Kazakhstan

INFOMATRIX ASIA 2026

## PROBLEM

**Variants of Uncertain Significance (VUS)** in DNA repair genes prevent accurate cancer risk assessment for patients and clinicians.

Existing predictors (REVEL, CADD) have **limited coverage** and poor performance on **non-BRCA2 HR genes** — PALB2, RAD51C, RAD51D.

**Can an ensemble ML model accurately classify missense variants across 5 HR genes?**

## DATASET

**19,223 variants** across 5 genes from ClinVar + gnomAD v4

BRCA2 · BRCA1 · PALB2 · RAD51C · RAD51D

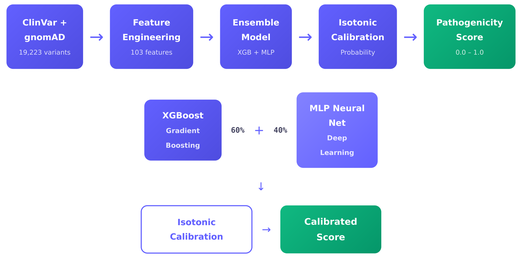
gnomAD proxy-benign variants (485) added to fix class imbalance in rare genes

## 103 ENGINEERED FEATURES

| | |
|---|---|
| 22 ESM-2 Embeddings | 42 Amino Acid Encoding |
| 13 Biochemical | 10 Structural (AlphaFold) |
| 4 Conservation | 4 Interaction Terms |
| 3 AlphaMissense | 2 Positional |
| 2 SpliceAI | 1 MAVE Score |

Gene-identifying features intentionally excluded to force transferable learning

## METHODOLOGY




10-fold stratified cross-validation (gene × label stratified) · SMOTE oversampling · Early stopping

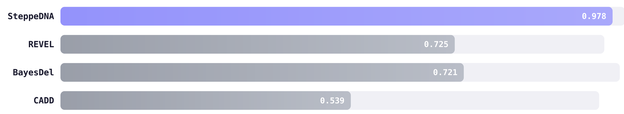
## RESULTS

| 0.978 | 0.881 | 95.7% |
|---|---|---|
| ROC-AUC | MCC | SENSITIVITY |

| GENE | TEST N | ROC-AUC | MCC | BAL. ACC. | TIER |
|---|---|---|---|---|---|
| **BRCA2** | 2,017 | 0.983 | 0.891 | 96.9% | High |
| **RAD51D** | 82 | 0.804 | 0.437 | 72.0% | Moderate |
| **RAD51C** | 135 | 0.743 | 0.403 | 70.6% | Moderate |
| **BRCA1** | 1,087 | 0.706 | 0.312 | 66.0% | Low |
| **PALB2** | 524 | 0.641 | 0.346 | 65.1% | Low |

10-Fold CV AUC: 0.9797 ± 0.0031 · Macro-average AUC: 0.775 · Test set n = 3,845

## VS. STATE-OF-THE-ART




Comparison on SteppeDNA's held-out test set (ROC-AUC). Competitors have 72% coverage vs. 100% for SteppeDNA.

## KEY INNOVATION

**gnomAD proxy-benign augmentation** dramatically improved non-BRCA2 gene performance:

| RAD51C | RAD51D |
|---|---|
| 0.509 → 0.743 | 0.586 → 0.804 |
| +0.234 | +0.218 |

| BRCA1 | PALB2 |
|---|---|
| 0.563 → 0.706 | 0.546 → 0.641 |
| +0.143 | +0.095 |

485 common population variants corrected extreme class imbalance (>90% pathogenic) in non-BRCA2 genes

## CONCLUSIONS

1. SteppeDNA achieves **AUC 0.978** across 5 HR genes with **100% variant coverage**
2. Population-frequency augmentation is **critical** for data-scarce genes
3. Per-gene reliability tiers ensure **honest, transparent** reporting
4. Web platform delivers **SHAP explanations**, ACMG evidence, and **3D protein visualization**

**Future work:** Larger ESM-2 models (650M), prospective clinical validation, Central Asian population data